Lecture 2.2: Arrays Basics and Common Methods

Instructor note: This file uses inline styles similar to the template. Consider linking to notes-template.css for consistent site-wide styling. Also the filename suggests "Lecture-1-10" while the heading says "Lecture 2.2" — consider making these consistent.

What is an array?

An **array** stores multiple values in a single variable. Elements are accessed using indexes that start at **0**.

```
var numbers = [2, 5, 2, 6, 2, 3];
console.log(numbers[0]); // 2
```

Output:

```
2
```

Printing arrays

You printed arrays in two ways:

```
console.log(numbers);           // shows array in console
console.log(numbers.toString()); // converts to comma-separated string
```

Output:

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```
[2, 5, 2, 6, 2, 3]
2,5,2,6,2,3
```

Array length

```
console.log(numbers.length);
```

Output:

```
6
```

Searching: `indexOf`

`indexOf(value)` returns the first index where the value occurs, or `-1` if not found.

```
console.log(numbers.indexOf(2));      // first 2
console.log(numbers.indexOf(2, 1));   // search from index 1 (helps find second occurrence)
```

Output:

```
0
2
```


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Adding/removing at the end: `push` and `pop`

```
numbers.push(85); // adds to end
numbers.pop();    // removes from end and returns the removed value
```

Output:

```
After pushing 85: 2,5,2,6,2,3,85
After popping 85 (returned value): 85
Array after pop: 2,5,2,6,2,3
```

Insert/remove anywhere: `splice`

`splice(startIndex, deleteCount, ...itemsToInsert)`

```
// Insert 45, 65, 75 at index 3 (deleteCount = 0 means delete nothing)
numbers.splice(3, 0, 45, 65, 75);

// Remove 2 items starting from index 4
numbers.splice(4, 2);
```

Output:

```
After inserting at index 3: 2,5,2,45,65,75,6,2,3
After removing 2 items from index 4: 2,5,2,45,6,2,3
```

> **Important:** `splice` changes (mutates) the original array.

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Sorting and reversing

```
numbers.sort();
numbers.reverse();
```

Output:

```
After sort() (string sort): 2,2,2,3,45,5,6
After reverse(): 6,5,45,3,2,2,2
```

Instructor insight: numeric sort needs a compare function

By default, `sort()` sorts items as **strings**. That means:

```
[2, 10, 3].sort(); // becomes [10, 2, 3] (string comparison)
```

Output:

```
10,2,3
```

For real numeric sorting:

```
numbers.sort((a, b) => a - b); // ascending
numbers.sort((a, b) => b - a); // descending
```

Output:

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```
// Example with [2,10,3]
Ascending numeric sort: 2,3,10
Descending numeric sort: 10,3,2
```

Other methods mentioned

Method	What it does
`shift()`	Remove from the beginning
`unshift(x)`	Add to the beginning
`slice(start, end)`	Returns a portion of the array (does NOT modify original)

> **Instructor insight / accuracy checks:** A few common points students mix up — `shift()` removes the first element, `unshift(x)` adds one or more elements to the beginning. `splice()` mutates the original array (inserts/removes), while `slice(start, end)` returns a new array and does not modify the original. `indexOf()` uses strict equality (===) for comparisons. Also, `sort()` without a compare function sorts items as strings; provide a numeric compare for numeric ordering.

Practice tasks

1. Create an array of 6 numbers and print it.
2. Find the index of a value; if it's `-1`, print "Not found".
3. Insert 3 new items in the middle using `splice`.
4. Sort the array numerically (ascending), then reverse it.

JS Refined Notes — Lecture 2.2